Traces of neutral hydrogen at high redshift absorb flux just blueward of the Lyα resonance. The resulting Gunn–Peterson[28] trough has been observed in the spectrum of the highest redshift quasar[7,29], and is the primary indicator of the reionization epoch. We find that about 40% of multiple image lens galaxies ($i^* < 22.2$) contribute flux in the Gunn–Peterson trough above the current upper limit. Contamination of the Gunn–Peterson trough in lensed quasar spectra will therefore limit their use as a probe of the epoch of reionization.

The high source redshifts imply image separations that are slightly larger than usual[16], about 1–2″. In addition, the lenses are likely to be found at higher redshift[16] owing in part to the higher source redshifts but also because bright (low-z) lenses are excluded. Lensing of high-redshift quasars allows measurement of the masses of the lens galaxies[21] at redshifts higher than is currently possible. Furthermore, quasar microlensing should be common over a ten-year base line[26]. This offers the exciting possibility of measuring the source size, and hence black hole mass, which in turn yields the ratio between quasar luminosity and its Eddington value, a quantity that will be useful in constraining models of structure formation[3]. □

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The nature and transport mechanism of hydrated hydroxide ions in aqueous solution

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Compared to other ions, protons (H^+) and hydroxide ions (OH^-) exhibit anomalously high mobilities in aqueous solutions[1]. On a qualitative level, this behaviour has long been explained by 'structural diffusion'—the continuous interconversion between hydration complexes driven by fluctuations in the solvation shell of the hydrated ions. Detailed investigations have led to a clear understanding of the proton transport mechanism at the molecular level[2–8]. In contrast, hydroxide ion mobility in basic solutions has received far less attention[2,3,9,10], even though bases and base catalysis play important roles in many organic and biochemical reactions and in the chemical industry. The reason for this may be attributed to the century-old notion[11] that a hydrated OH^- can be regarded as a water molecule missing a proton, and that the transport mechanism of such a 'proton hole' can be inferred from that of an excess proton by simply reversing hydrogen bond polarities[11–18]. However, recent studies[2,3] have identified OH^- hydration complexes that bear little structural similarity to proton hydration complexes. Here we report the solution structures and transport mechanisms of hydrated hydroxide, which we obtained from first-principles computer simulations that explicitly treat quantum and thermal fluctuations of all nuclei[19–21]. We find that the transport mechanism, which differs significantly from the proton hole picture, involves an interplay between the previously identified hydration complexes[2,3] and is strongly influenced by nuclear quantum effects.

Transport of a hydrated proton, H_3O^+, in water involves a continuous interconversion between two hydration complexes (Fig. 1a–c). A tricoordinate $H_3O^+(H_2O)_3$ complex (Fig. 1a) is transformed via proton transfer into a $[H_2O\cdots H\cdots OH_2]^+$ or $H_5O_2^+$ complex (Fig. 1b). This process, known as the 'structural diffusion' or 'Grotthuss' mechanism[1], is driven by fluctuations in the second solvation shell of H_3O^+, which reduce the coordination of a first-solvation-shell water from four to three[2–5]. This water molecule is, thus, 'prepared' to become a properly solvated H_3O^+ by proton transfer $H_3O_a^+\cdots H_2O_b \rightleftharpoons H_2O_a\cdots H_3O_b^+$, where O_a and C_b are the two oxygens involved in the hydrogen bond, respectively.

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Owing to quantum effects, the actual defect structure is best described as a fluxional defect[5]. Applying the proton-hole concept to the OH^- case, one would expect the occurrence of tricoordinate $OH^-(H_2O)_3$ (in which the hydroxide oxygen accepts three hydrogen bonds) and $H_3O_2^-$ (that is $[HO\cdots H\cdots OH]^-$) complexes, analogous to $H_3O^+(H_2O)_3$ and $H_5O_2^+$, respectively. Moreover, structural diffusion would be driven by similar second-solvation-shell fluctuations. Interpretations of Raman and infrared spectra of concentrated basic solutions have been guided by the proton-hole picture, a fact that has led to controversies about the structure and dynamics of hydrated hydroxide[14–17]. Prompted by this controversy, we undertook an extensive *ab initio* path integral[19,20] study of a hydroxide ion in bulk water at room temperature (see Fig. 2 for details).

For each configuration generated in the simulation, the OH^- charge defect, designated $(O^*H')^-$, is located in the hydrogen-bond (H-bond) network by identifying the oxygen with a single hydrogen. Next, for each H-bond, a displacement coordinate, $\tilde{\delta} = R_{O_aH} - R_{O_bH}$, is defined, where $R_{O_a}H$ and $R_{O_b}H$ are the distances between the shared proton and the two oxygens. A small value of $\tilde{\delta}$ indicates a propensity for proton transfer. Finally, a transfer coordinate, δ, is defined in each configuration by selecting the H-bond with the smallest $\tilde{\delta}$. This H-bond, designated $O^*\cdots H^*\bar{O}$, is considered to be the 'most active' or most likely to experience proton transfer[5]. Invariably, it is found that one of the oxygens in this H-bond corresponds to the defect, O^*.

In order to identify the role played by various solvation complexes of OH^-, configurations with large and small $|\delta|$ are selected. Figure 2a shows that, at large $|\delta|$, O^* accepts, on average, four H-bonds from neighbouring waters. Representative configurations

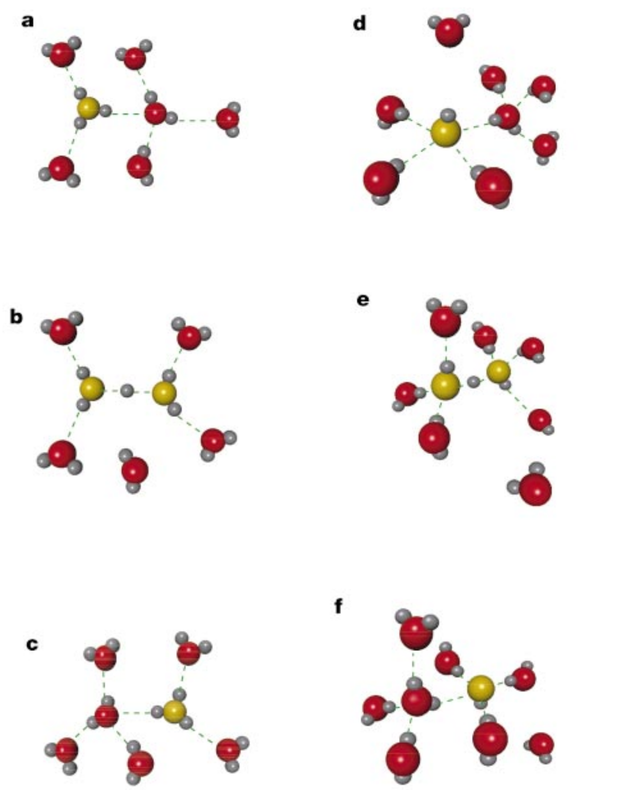

Figure 1 Schematic illustration of proton and hydroxide ion transport in water at room temperature. Panels **a**–**c** show the transport mechanisms of H_3O^+, and panels **d**–**f** the transport mechanisms of OH^-. Red and grey spheres are oxygen and hydrogen atoms, respectively, and yellow spheres are the H_3O^+ and OH^- defect oxygens. Dashed green lines denote hydrogen bonds. These simplified sketches are idealized renderings of the classical limit of the 'real' processes shown in Fig. 3 and Fig. 1 of ref. 5. H_3O^+ mechanism: In **a**, the H_3O^+ is shown in a threefold-coordinated, $H_3O^+(H_2O)_3$, state, and a complete coordination shell is shown for one of the first-solvation-shell waters. Thermal fluctuations cause an H-bond breaking between a first and second solvation-shell water of H_3O^+ to occur (**b**), leaving the first-solvation-shell water with a threefold coordination pattern, followed by a shrinking of the first-shell O*O distance and a migration of the proton to the centre of the bond, forming an intermediate $H_5O_2^+$ complex. Complete transfer of the proton (**c**), yields a properly solvated H_3O^+ at a new site in the H-bond network. The 'gate closes' (no further proton transfer events are possible until another H-bond breaking event occurs) if the newly formed water acquires a fourth H-bond (shown as the water released by the H-bond breaking event in **b**, although it could be any nearby solvent water). OH^- mechanism: In **d**, the OH^- is shown in a fourfold-coordinated, $OH^-(H_2O)_4$, state, and a complete coordinate shell is shown for one of the first-solvation-shell waters. A neighbouring solvent is shown above the OH^- hydrogen. Thermal fluctuations cause a first-solvation-shell H-bond breaking event (**b**), leaving the OH^- in a threefold $OH^-(H_2O)_3$ state, followed by a shrinking of a first-shell O*O distance and the formation of a weak H-bond between the OH^- hydrogen and a neighbouring solvent water (see, also, Fig. 3b–d). Complete transfer of the proton (**f**) causes the $OH^-(H_2O)_3$ to migrate to a new site in the H-bond network. The 'gate closes' if the newly formed hydroxide acquires a fourth H-bond (shown as the water released in **b**, although it could be any nearby solvent water), to complete the $OH^-(H_2O)_4$ structure.

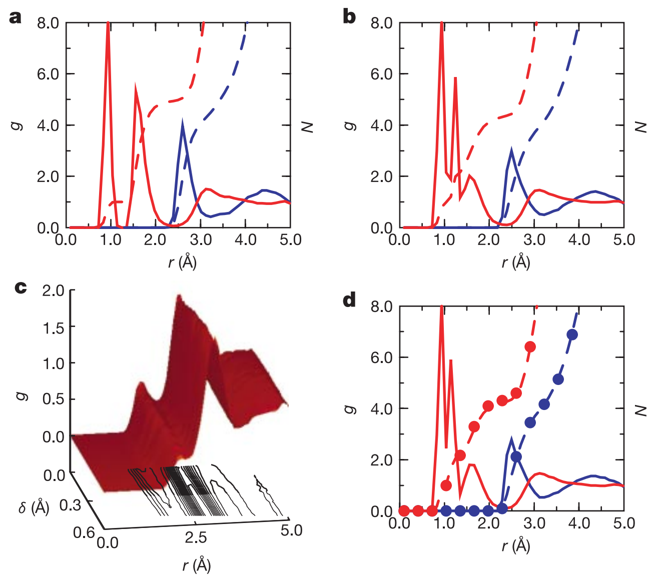

Figure 2 Radial distribution functions and coordination numbers during proton transfer. Shown are radial distribution functions, $g(r)$, and, on the same scale, running coordination numbers, $N(r)$, with respect to the hydroxide ion and a first-solvation-shell water at different stages in the proton transfer process. **a**, Radial distribution functions, g_{O^*O} (solid blue) and g_{O^*H} (solid red) and coordination numbers, N_{O^*O} (dashed blue) and N_{O^*H} (dashed red), with respect to the OH^-, O^*, for values of the proton transfer coordinate $|\delta| = R_{O^*H^*} - R_{\bar{O}H^*} > 0.5$ Å. Here, H^* denotes the shared proton, and $\bar{O}$ denotes the oxygen of a water molecule that donates the 'most active' H-bond to O^*, that is, $O^*\cdots H^*\bar{O}$. **b**, As **a** except for $|\delta| < 0.1$ Å. **c**, Two-dimensional generalization of a hydrogen-oxygen radial distribution function $g_{H'O}(r,|\delta|)$ with respect to the OH^- hydrogen, H', as a function of r and $|\delta|$. **d**, Radial distribution functions and coordination numbers with respect to $\bar{O}$ for $|\delta| < 0.1$ Å. The colour and line types are the same as in **a**. Coordination numbers from **b** are superimposed with red (O*H) and blue (O*O) circles. The simulations were performed using CPMD Version 3.0 (J. Hutter *et al.*, Max-Planck-Institut für Festkörperforschung and IBM Zurich Research Laboratory 1995–1996)[21] at ambient conditions using 31 H_2O with one OH^- anion in a periodic cubic box of side length 9.8692 Å, employing the BLYP functional, Troullier-Martins type pseudopotentials, and a Γ-point plane wave expansion of the valence orbitals up to 70 Ry. This particular scheme has been shown to yield a good description of dynamical processes in aqueous systems[5,6,30,32]. The path integral was discretized using $P = 8$ Trotter replicas following the methodology of ref. 20. After careful equilibration, quantum and classical nuclear trajectories consisting of 322,000 and 346,000 configurations, with time steps of 5.0 a.u. and 7.0 a.u., respectively, were generated via the Car-Parrinello algorithm[22].

(Fig. 3a) reveal that the accepted H-bonds are in a roughly planar arrangement, forming the $OH^-(H_2O)_4$ complex. Previous solution phase[2] and quantum-chemical[23] calculations have confirmed the stability of this structure. Furthermore, a recent spectroscopic study indicates that there is no cationic analogue of this complex[24]. Importantly, it is also found that in roughly half of the configurations, the hydroxide hydrogen, H′, donates an additional weak H-bond, giving a total O*O coordination number of $N_{O^*O} \approx 4.5$.

For small $|\delta|$, the O*O peak in the radial distribution functions (Fig. 2b) shifts to smaller distances, and N_{O^*O} decreases from 4.5 to

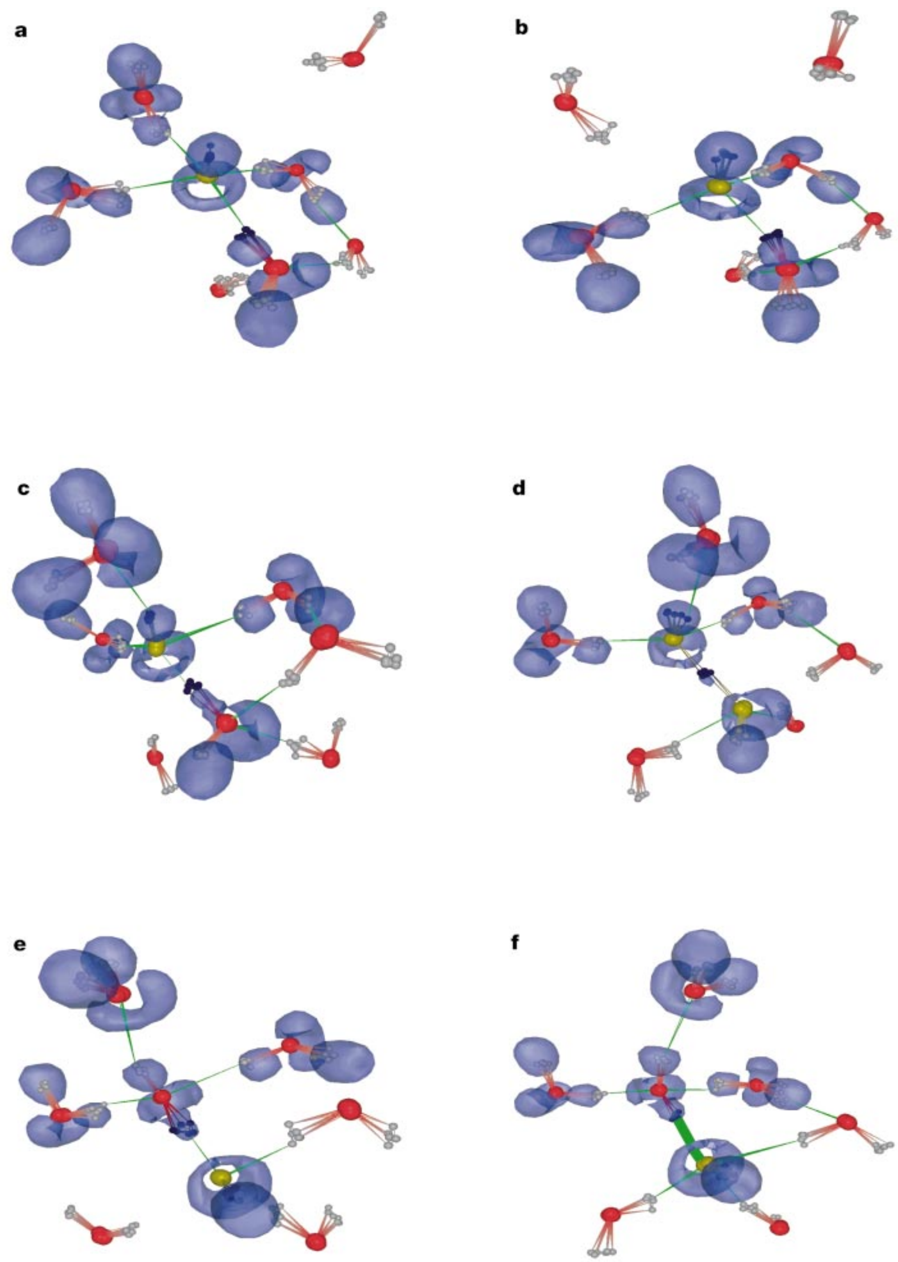

Figure 3 Representative configurations showing the proton transfer mechanism. Each panel shows quantum-mechanical particle density snapshots of the OH^- together with relevant first and second solvation-shell water molecules. Oxygen and hydrogen atoms are shown as red and grey spheres, respectively. The defect oxygen, O* and shared proton in the most active H-bond are shaded yellow and black, respectively. H-bonds are schematically shown as green lines. In blue are shown the $\eta = 0.93$ isosurfaces of the electron localization function (ELF)[29] of the path centroid configuration for the OH^- and only those water molecules which are H-bonded to the OH^-. **a**, The OH^- is in its inactive $OH^-(H_2O)_4$ structure with fourfold planar coordination around O*. The ELF shows that the three lone pairs around O* are in a delocalized ring structure, thus supporting the hypercoordination of the OH^- (see text). An additional water molecule, which participates in the activation process (**c** and **d**), appears in the upper right corner. **b**, A first-solvation-shell H-bond breaking event occurs, which transforms the $OH^-(H_2O)_4$ structure into an approximately tetrahedral $OH^-(H_2O)_3$ structure. **c**, A weak H-bond between the OH^- hydrogen, H′, and the bulk water molecule from **a** and **b** is formed. As shown, when this H-bond forms, one of the shared protons between O* and a coordinating water begins to transfer. **d**, As the proton is transferred, the environments around each OH moiety become similar (see, also, Fig. 2d). (Note that a complete solvation shell for the water that transfers its proton to O* is not shown. However, the coordination number around this water is, nevertheless, 4.0—see Fig. 2d). The ELF shows two distorted rings around the yellow oxygens, emphasizing the symmetry of this configuration. **e**, After the proton transfer, the original OH^- has been transformed into a properly solvated water molecule with two lone pairs, whereas the newly formed OH^- (now shown in yellow) possesses the ring-shaped ELF. **f**, The process is completed by the acceptance of a fourth H-bond from the newly formed O*, which ‘closes the gate’ by forming a new, inactive $OH^-(H_2O)_4$ complex with a fully intact ELF ring as in **a**. Although *ab initio* path-integral molecular dynamics method does not rigorously produce real-time evolution, processes that must occur and their approximate sequence can be inferred qualitatively.

4.1, very close to the value, 4.0, obtained for bulk water molecules. The number of accepted and donated H-bonds is 3.2 and 0.9, respectively. These imply that the emergent structure during proton transfer is the tetrahedral $OH^-(H_2O)_3$ complex in which O* is approximately threefold coordinated (Fig. 3b), and that the donated H-bond involving H′ plays a crucial role in activating the proton transfer process (Fig. 3c, d). The importance of this H′···O H-bond is further demonstrated in the two-dimensional radial distribution function, $g_{H'O}(r,|\delta|)$ of Fig. 2c. A weak signal at $r \approx 2$ Å, first apparent at large $|\delta|$, becomes prominent as $|\delta| \rightarrow 0$. It should be noted that the location of the O*O peak of the δ-averaged radial distribution function, 2.5 Å, is in good agreement with recent neutron diffraction data[25]. Moreover the δ-averaged O*O coordination, 4.3, is consistent with experimental estimates[26] (although the "effective hydration number" in ref. 26 is not precisely equivalent to the microscopic coordination number).

The above findings have led us to propose an alternative mechanism to that of the proton-hole picture (Fig. 3). Activation of a proton transfer event requires that the solvation pattern around OH^- closely resemble that of a bulk water molecule. This pattern (which is not a free-energy minimum), is realized by two concerted yet distinct processes. The first is a transformation from the inactive $OH^-(H_2O)_4$ to the active $OH^-(H_2O)_3$ solvation state, a process that involves the breaking of a H-bond in the first solvation shell of OH^- (as shown in the change from Fig. 3a to Fig. 3b). The relatively facile conversion from $OH^-(H_2O)_4$ to $OH^-(H_2O)_3$ is consistent with the estimated activation energy, 1.16 kcal mol^{-1}, reported in ref. 23 and with recent spectroscopic data[24]. The importance of this step can be understood from the chemical differences between the $OH^-(H_2O)_4$ and $OH^-(H_2O)_3$ complexes. In the former, proton transfer through a H-bond would transform the OH^- defect into a H_2O molecule that accepts three H-bonds, an unfavourable configuration. In the latter, proton transfer through one of the acceptor H-bonds produces a H_2O with the correct coordination. The second process required to initiate proton migration is the formation of a weak H-bond between the hydroxide hydrogen, H′, and a neighbouring water molecule (Fig. 3b to c). When this occurs in the $OH^-(H_2O)_3$ complex, the OH^- is (fourfold) coordinated in such a way that when the proton, H*, is transferred, the original $(O^*H')^-$ becomes a tetrahedrally solvated water molecule with an ideal bulk solvation pattern. Note that, in the proton-hole picture, this H′···O H-bond (Figs 2c, 3c, 3d) is assumed never to form.

Once the $OH^-(H_2O)_3$ complex is activated, the environment around the two oxygens in the most active H-bond is nearly identical (Fig. 2d). This observation suggests that, for $|\delta| \approx 0$, the structural motif of a solvated $H_3O_2^-$ core, emerges. However, in contrast to the proton-hole scenario, here, the $H_3O_2^-$ is a transient state that is necessarily visited when the proton is transferred. Thus, the complete structural diffusion mechanism, depicted in Fig. 3, can be summarized as: (1) transformation of $OH^-(H_2O)_4$ to $OH^-(H_2O)_3$ by first-solvation-shell H-bond breaking (Fig. 3a to b); (2) activation of $OH^-(H_2O)_3$ by formation of the H′···O H-bond (Fig. 3b to c); (3) formation of the transient $H_3O_2^-$ complex by partial proton transfer (Fig. 3c to d); (4) completion of proton transfer resulting in a new O* and $OH^-(H_2O)_3$ complex (Fig. 3d to e); (5) acceptance of a fourth H-bond by the new O* resulting in a change from $OH^-(H_2O)_3$ to $OH^-(H_2O)_4$ and consequent inactivation (Fig. 3e to f). This completes the cycle: the $OH^-(H_2O)_4$ complexes in Figs 3f and 3a are isostructural but are located at different sites in the network. In Fig. 1d–f a simplified sketch of this mechanism is contrasted to that for structural diffusion of H_3O^+ in acids (Fig. 1a–c).

Having established the mechanism, we examine how closely a purely classical treatment of the nuclear motion at 300 K approximates this picture. Classically (Fig. 4a), there is a significant funnelling of the angle (θ) between the O*H′ and O*Ō bonds about the HOH bulk water bend angle (~104°) for $|\delta| \approx 0$, implying that the $OH^-(H_2O)_3$ complex must be close to its ideal tetrahedral geometry. However, when nuclear quantum effects are included (Fig. 4b), this funnelling is largely washed out, implying a significant probability of proton transfer even when the geometry of the complex is distorted. Such a pronounced 'corner cutting' quantum effect[27], having no analogue in hydrated H_3O^+ (ref. 5), may also explain why the observed H/D isotope effect on the mobility is larger in basic than in acidic solutions[18].

Nuclear quantum effects also lower the proton transfer free energy barrier along the δ-coordinate from roughly 1.28 to 0.34 kcal mol^{-1} (Fig. 4c). Despite this reduction, the free energy retains its double-well character, implying that $H_3O_2^-$ is a transient complex or activated state (Figs 2d and 3d). This contradicts the proton-hole picture, in which $H_3O_2^-$ is understood to be a stable complex. It is also in contrast to the hydrated H_3O^+ case, in which a smaller (0.6 kcal mol^{-1}) barrier obtained at the classical level is washed out by nuclear quantum effects[5]. Finally, it contrasts with the gas-phase studies in ref. 28, where a pronounced barrier in the $H_3O_2^-$ complex is also completely washed out by nuclear quantum effects. If the present proton transfer free-energy value, 0.34 kcal mol^{-1}, is combined with the 1.16 kcal mol^{-1} (ref. 23)

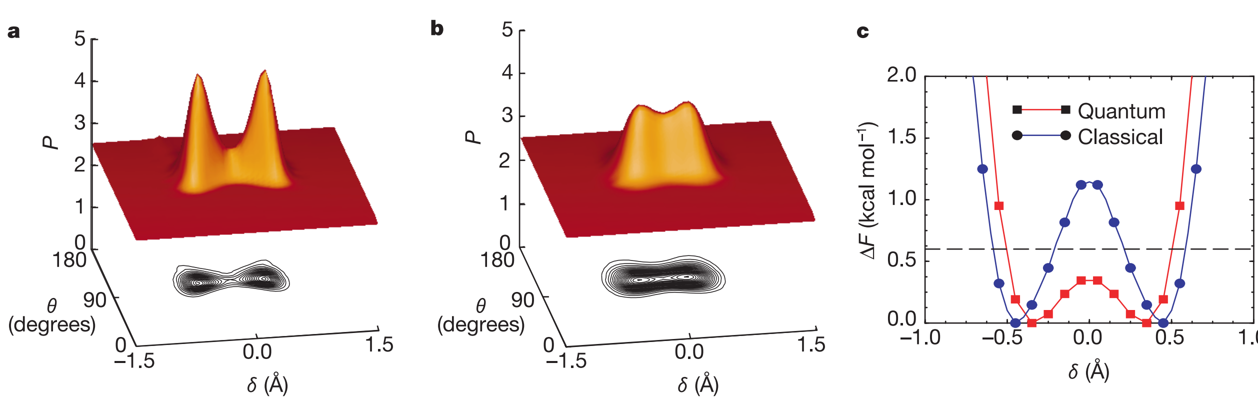

Figure 4 Structural changes occurring during proton transfer. Shown are the normalized two-dimensional probability distribution function, $P(\delta, \theta)$, of the proton displacement coordinate, δ and the angle, θ, between the OH^- covalent bond axis and the O*–Ō vector of the most active H-bond for the classical (**a**) and quantum (**b**) simulations at 300 K. Contours of these distribution functions are also shown. The distribution functions are smoothed for presentation and symmetrized about $\delta = 0$ Å. Comparison of the free-energy profiles (**c**) along the δ coordinate for classical (circles) and quantum (squares) treatment of the nuclei. Note the centroid of the coordinate, δ, has been used for the quantum profile.

required for step (1) of the above mechanism, and assuming an additional 0.5 kcal mol^{-1} (ref. 18) to reduce the length of each of the three H-bonds in the $OH^-(H_2O)_3$ complex by 0.1 Å after step (1), an estimate of approximately 3 kcal mol^{-1} is obtained for the total activation energy of our mechanism. This rough estimate is consistent with the experimental value obtained from hydroxide mobility data[18].

The unexpectedly high coordination of the solvated hydroxide[2,23,26] contrasts with the textbook picture that each of the three lone pairs of O* accepts a single H-bond. An explanation of this phenomenon is obtained by analysing the electron localization function[29] along the mechanistic path of Fig. 3. The electron localization function indicates spatial regions where electron pairs are most likely to be found. The most striking feature of these functions (Fig. 3) is the presence of a continuous ring around the OH^- bond axis[30,31]. This finding is corroborated by its electrostatic potential that displays a ring of maximum negative charge around its O terminus. This pattern, which suggests a lack of directionality in H-bonding to O*, supports both structural flexibility and high coordination, and is consistent with the spectroscopic picture of ref. 24.

The analysis presented here emphasizes the complex chemical differences between the hydrated H_3O^+ and OH^- ions, and their effect on the anomalous transport mechanisms in acidic and basic solution (Fig. 1). We recognize the temptation to invoke symmetry arguments as a method to intuit mechanisms of seemingly similar processes. However, this approach cannot yield a correct picture when the chemistry of the two species is sufficiently different. Nevertheless, a common pattern of charge migration in acids and bases emerges: both require that the nascent species in the proton transfer process be correctly 'pre-solvated' by way of specific fluctuations in the local H-bond network. □

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Nonlinear dynamics of ice-wedge networks and resulting sensitivity to severe cooling events

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Patterns of subsurface wedges of ice that form along cooling-induced tension fractures, expressed at the ground surface by ridges or troughs spaced 10–30 m apart, are ubiquitous in polar lowlands[1]. Fossilized ice wedges, which are widespread at lower latitudes, have been used to infer the duration[2–4] and mean temperature[5,6] of cold periods within Proterozoic[2] and Quaternary climates[3–13], and recent climate trends have been inferred from fracture frequency in active ice wedges[14]. Here we present simulations from a numerical model for the evolution of ice-wedge networks over a range of climate scenarios, based on the interactions between thermal tensile stress, fracture and ice wedges. We find that short-lived periods of severe cooling permanently alter the spacing between ice wedges as well as their fracture frequency. This affects the rate at which the widths of ice wedges increase as well as the network's response to subsequent climate change. We conclude that wedge spacing and width in ice-wedge networks mainly reflect infrequent episodes of rapidly falling ground temperatures rather than mean conditions.

Ice wedges originate with fractures opened in perennially frozen ground by tensile stress induced by falling winter temperatures[15–17].

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